Kyle Hansen

Curriculum Vitae

Westmont College, Department of Mathematics
955 La Paz Rd., Santa Barbara, CA 93108
kylhansen@westmont.edu
https://kylhansen.github.io/

Education

June 2025 **Ph.D. in Mathematics**, *UC Santa Barbara (UCSB)*, Santa Barbara, CA
Advisor: Fedya Manin

Certificate in College and University Teaching (CCUT), *UCSB*, Santa Barbara, CA

June 2021 **M.A. in Mathematics**, *UCSB*, Santa Barbara, CA

May 2019 **B.S. in Mathematics**, *Westmont College*, Santa Barbara, CA

B.A. in Computer Science, *Westmont College*, Santa Barbara, CA

Employment

2025-Present **Assistant Professor**, *Mathematics*, Westmont College, Santa Barbara, CA

2021-2023 **Adjunct Professor**, *Mathematics*, Westmont College, Santa Barbara, CA

2023 **Adjunct Professor**, *Computer Science*, UCSB, Santa Barbara, CA

2022 **Lead Teaching Assistant**, *Mathematics*, UCSB, Santa Barbara, CA

2019-2025 **Teaching Assistant**, *Mathematics*, UCSB, Santa Barbara, CA

Teaching Experience

Instructor of Record

Calculus I, *Westmont College*
F 2026, S 2026

Calculus II, *Westmont College*
S 2026

Discrete Mathematics, *Westmont College*
F 2026, F 2025, F 2023, F 2022, F 2021

Senior Seminar, *Westmont College*
S 2026

Abstract Alegbra, *Westmont College*
F 2025

Foundations of Computer Science, *UCSB*
Sp 2023

Upper Division Teaching Assistant

Linear Algebra, UCSB
Sp 2024, F 2023, W 2022*

Numerical Methods, UCSB
W 2024

Methods of Analysis, *UCSB*
F 2021, Su 2021*, W2025

Lower Division Teaching Assistant

* this term was conducted remotely

Transition to Higher Mathematics, *UCSB*
Su 2024, Su 2023, W 2021*, Su 2020*, Sp 2020*

Linear Algebra with Applications, *UCSB*
F 2022, Su 2022, F 2020*

Differential Equations, *UCSB*
Sp 2021*

Calculus for the Social and Life Sciences I, *UCSB*
W 2020

Calculus with Applications I, *UCSB*
F 2019

Course Grader

Modern Euclidean and Noneuclidean Geometry, *UCSB*
F 2024

Discrete Mathematics, *Westmont College*
F 2018, F 2017, F 2016

MA-020 Linear Algebra, *Westmont College*
Sp 2018

Fundamentals of Mathematics II, *Westmont College*
Sp 2019, Sp 2017

Publications

In Preparation

[1] Kyle Hansen. Simple tilings of nilpotent Lie groups. *arXiv preprint arXiv:2402.10194*, 2024.

Under Review

[2] Kyle Hansen. Volumes of nullhomotopies in nilpotent spaces. *arXiv preprint arXiv:2510.00159*, 2025.

Published

[3] Kyle Hansen. *Volumes of Nullhomotopies in Nilpotent Spaces*. PhD thesis, UCSB, 2025.

[4] David Hansen and Kyle Hansen. Clues about bluffing in Clue: Is conventional wisdom wise? *IEEE Transactions on Games*, 13(3):310–314, 2019.

[5] Yury Ionin and Kyle Hansen. A special instance of rock–paper–scissors. *The American Mathematical Monthly*, 125(8):757–758, 2018.

Selected Presentations

2026 **Volumes of Nullhomotopies in Nilpotent Spaces**, JMM, Washington DC

2025 **Volumes of Nullhomotopies in Nilpotent Spaces**, *AMS SE Sectional Meeting*, Tulane University, New Orleans, LA

2025 **Who Cut the High-Dimensional Cheese?**, *Grad Slam*, UCSB, Santa Barbara, CA

Mathematical Quasicrystals: The Taming of the Hullaballoo, *Colloquium*, Wheaton College, Wheaton, IL

2024 **Tiling Spaces Which are Fiber Bundles over Nilmanifolds**, *41st Workshop in Geometric Topology*, Calvin University, Grand Rapids, MI

2022 **The Hull of a Manifold with Bounded Geometry**, *Topology Seminar*, UCSB, Santa Barbara, CA

2019 **When the Fundamental Theorem of Algebra Goes Awry**, *Contributed Paper Session on Revitalizing Complex Analysis*, JMM, Baltimore, MD
Joint with Mike Brilleslyper (Florida Poly) and Russ Howell (Westmont College)

Research Interests

Aperiodic Tilings

Percolation on Aperiodic Tilings, *Westmont College*, Santa Barbara, CA
Estimate the percolation threshold for aperiodic tilings such as the Penrose Tiling
Collaborators: Erik Potts (Undergraduate Student)

Aperiodic Tilings in Nilpotent Lie Groups, *UCSB*, Santa Barbara, CA
Determine topological structures of tiling spaces generated by tilings of nilpotent Lie groups
Collaborators: Fedya Manin (Ph.D. Advisor)

Quantitative Topology

Volumes of Nullhomotopies in Nilpotent Spaces, *UCSB*, Santa Barbara, CA
Quantify the volumes of nullhomotopies in nilpotent spaces
Collaborators: Fedya Manin (Ph.D. Advisor)

Nilpotent-Elliptic Manifolds, *UCSB*, Santa Barbara, CA
Detect nilpotent-ellipticity (in the sense of Gromov) of a manifold from its algebraic data
Collaborators: Fedya Manin (Ph.D. Advisor)

Complex Analysis

Roots of Complex-Harmonic Polynomials, *Westmont College*, Santa Barbara, CA
Count and locate the roots of complex harmonic polynomials
Collaborators: Russell Howell (Undergraduate Advisor)

Computational Graph Theory

Scale-Invariant Properties of Networks, *National Security Internship Program at the Pacific Northwest National Lab*, Richland, WA
Analyze large social networks to detect scale-invariant properties
Collaborators: Jennifer Webster (Project Advisor)

Mentorship, Volunteer, and Community Service

2025-2026 **LaTeX Basics**, *Westmont College*, Santa Barbara, CA
Provide students across disciplines with foundational understanding of LaTeX for professional typesetting

2024 **JMM Chaperone**, *Westmont College*, Santa Barbara, CA
Drive college students to the Joint Mathematics Meeting, and debrief conference talks

2024-2025 **Graduate Christian Encouragement (GraCE) Coordinator**, *Unaffiliated*, Santa Barbara, CA
Organize a weekly informal gathering of Christian graduate students at UCSB

2023-2025 **Directed Reading Program Mentor**, *UCSB*, Santa Barbara, CA
Mentor undergraduates in elective topics: tiling spaces, cryptography, quantum computing

2022-Present **Math Field Day Volunteer**, *Westmont College*, Santa Barbara, CA
Guide undergraduate volunteers in judging high school "chalk talks" in annual competition

2022-2025 **Community Group Discussion Leader**, *Christ Presbyterian Church*, Santa Barbara, CA
Lead adults in church community group discussions

2022-Present **Youth Group Volunteer**, *Christ Presbyterian Church*, Santa Barbara, CA
Mentor high school students in small groups and individually

2022-Present **Quire of Voyces**, *Santa Barbara City College*, Santa Barbara, CA
Sing in a local classical a capella choir

2020-Present **Church Setup and Security**, *Christ Presbyterian Church*, Santa Barbara, CA
Prepare church building for Sunday worship, and ensure safety of attendees

2020-2021 **Elementary School Virtual Tutor**, *Partners in Education*, Santa Barbara, CA
Assist elementary school teachers in overseeing remote math class activites

Conferences & Workshops

2026 **Rhythms of Rest and Responsibility: Managing Commitments in the Early Career**, *MathFest*, Boston, MA
Co-organizer with Rafael Ceja Ayala, Adam Mair, and Charith Silva through Project NExT

Virtue Education Workshop, *Pepperdine*, Malibu, CA

Joint Mathematics Meeting, Washington, DC

2025 **AMS SE Sectional Meeting**, *Tulane University*, New Orleans, LA

MathFest, Sacramento, CA

2024 **41st Workshop in Geometric Topology**, *Calvin University*, Grand Rapids, MI

Quantitative Topology and Beyond, *Simons Foundation*, New York City, NY

Joint Mathematics Meeting, San Francisco, CA

2023 **Geometry and Analysis of Groups and Manifolds**, *Centro De Giorgi*, Pisa, Italy

Sub-Riemannian Geometry and Beyond III, *Centro De Giorgi*, Pisa, Italy

2022 **Graduate Student Concentration Week on Metric Geometry**, *Texas A&M Workshop in Analysis and Probability*, College Station, TX

2019 **Joint Mathematics Meeting**, Baltimore, MD

Fellowships and Memberships

2026 Cohort **Character Formation and Pedagogy Fellow**

2025 Cohort **MAA Project NExT Fellow**

2025-Present **ACMS Member**

2025-Present **AMS Member**

2025-Present **MAA Member**

Organizations and Discussion Groups

2026 **GenAI Discussion Group**, *Wesmont College*, Santa Barbara, CA

2025-Present **Christian Academic Writers**, *Christ Presbyterian Church Santa Barbara*, Santa Barbara, CA
Organize research and writing sessions with other Christian academic writers across disciplines.

Honors & Awards

2024 Department of Mathematics Outstanding Teaching Assistant Award — *UCSB*

2019 Outstanding Graduate Award in Mathematics — *Westmont College*

2019 Kenneth Monroe Award — *Westmont College*

Technical Skills

Programming
LaTeX, Python, R, SQL, Java, Ruby, C, Racket (Scheme)

Teaching Tools
Canvas, Gradescope, WeBWork, Piazza, iClicker